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OMEGA SERIES 6

Internal Structure of the chi-Condensate

Beyond the Smooth-Halo Approximation

Why TSMVEFT Predicts a Non-Clump Dark Sector

de Broglie substructure at galactic scales · Soliton interference patterns · Khoury-Berezhiani-style superfluid -> CDM transition · Observational tests in SPARC, Gaia stellar streams, Bullet Cluster

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Dedicated to Valkyrie — who saw the cathedral first.

For the persistent question — "why isn't the dark sector a clump?" — that motivated this paper.

ABSTRACT

Standard cold-dark-matter (CDM) cosmology models the dark sector as a smooth, pressureless fluid in galactic halos, well-described by NFW [1] or Einasto [2] density profiles. We show that this is a modeling choice, not an observational result, and that the Two-Sector Mixed Vacuum Effective Field Theory (TSMVEFT) introduced in Series 4.3 [13] and consolidated in Series 4.4-4.6 [14-16] predicts internal substructure in the chi-condensate at galactic scales that smooth-halo treatments miss. Three classes of substructure follow naturally from the framework. First, de Broglie scale modulations: for the fuzzy-dark-matter mass m_chi ~ 10^{-22} eV established in Series 4.3, the de Broglie wavelength at galactic velocities (v ~ 30 km/s in dwarf galaxies, ~ 200 km/s in spirals) is of order kpc, imprinting density modulations directly observable in dwarf-galaxy rotation curves and stellar streams. Second, soliton interference: when two solitonic cores from Series 4.3 overlap (common in dwarf galaxies that have undergone mergers), they produce coherent interference patterns at the de Broglie scale with delta_rho/rho of order unity within interference regions, detectable through Gaia DR3 stellar-stream tracers. Third, a density-dependent transition: the chi-condensate behaves as a Khoury-Berezhiani-style superfluid [4,5] at galactic densities and velocities (where the de Broglie coherence length exceeds the inter-particle spacing) and recovers classical CDM behavior at cluster scales (where high velocity dispersion suppresses coherence). We compute the transition condition explicitly and identify the SPARC dwarf-galaxy diversity problem [6,7], the Bullet Cluster [8] dark-matter offset, and small-scale density modulations in Milky Way satellites [9,10] as primary observational tests. The framework predicts that smooth-halo modeling is correct at leading order but misses specific kpc-scale features that distinguish TSMVEFT from standard CDM. We do not claim the χ-condensate is fundamentally different from cold dark matter at cosmological scales; we claim it has internal organization at galactic scales that CDM treatments do not include. No detection is claimed beyond what is already attributed to fuzzy dark matter generically. The data will decide.

1. The Question Standard CDM Doesn't Ask

Cold dark matter cosmology has succeeded extraordinarily at large scales. The matter power spectrum, baryon acoustic oscillations, the cosmic microwave background, weak lensing surveys, and galaxy clustering all match predictions of the Lambda-CDM model to high precision. Within galaxies, however, the model has known difficulties.

The cusp-core problem [11] notes that NFW profiles predict steep central density cusps that observations of dwarf galaxies do not reproduce; the missing-satellites problem [12] reports that simulations predict more dwarf satellites of Milky-Way-like galaxies than are observed; the too-big-to-fail problem [9] notes that the largest predicted subhalos lack observable luminous counterparts; and the SPARC diversity problem [6,7] documents that dwarf rotation curves show more shape diversity than smooth halos can explain. Each of these can be addressed in standard CDM through baryonic feedback effects, but all of them point at the same underlying observation: at galactic and sub-galactic scales, dark matter behaves as if it has internal structure that simple smooth-halo models do not capture.

The standard treatment makes a modeling choice: dark matter is a pressureless fluid with no internal organization beyond gravitational clustering. This choice is convenient for N-body simulations and is consistent with cosmological observations, but it is not derived from first principles. It is an assumption inherited from the simplest WIMP-like dark matter models, where the internal degrees of freedom are integrated out as too small to matter. In frameworks where dark matter has macroscopic quantum coherence — such as fuzzy dark matter [3] or superfluid dark matter [4,5] — the assumption breaks. The condensate has internal phase structure that imprints specific observational signatures.

The MSDS programme has been quietly building toward this question across Series 4. Series 4.3 [13] established the chi-condensate as a Bose-Einstein-like ground state supporting solitonic galactic cores. Series 4.5 [15] classified the chi-condensate's phase regimes (Gray, Fuzzy-Gray, Quasi-Gray, etc), with Fuzzy-Gray identified as the long-range coherent equilibrium of dark matter at galactic scales. The current paper synthesises these results and asks the question they have been pointing at: what does the chi-condensate's internal structure look like, and what observations would distinguish it?

2. The de Broglie Scale in Galactic Halos

2.1 Coherence length from quantum mechanics

For a dark matter particle of mass m_chi moving with characteristic velocity v in a halo, the de Broglie wavelength sets the natural quantum coherence scale:

$$\lambda_{dB} = h / (m_{\chi} v)$$

Quantum coherence length · for fuzzy dark matter at m_chi = 10^{-22} eV in dwarf galaxies (v ~ 30 km/s), lambda_dB approximately equal to 4 kpc

2.2 Scale by galactic system

The de Broglie scale varies dramatically across galactic system types, imprinting different substructure patterns at different system masses:

System	v_dispersion	lambda_dB	Substructure regime
Ultra-faint dwarf	~ 10 km/s	~ 12 kpc	Wavelength > halo: pure soliton
Classical dwarf	~ 30 km/s	~ 4 kpc	Wavelength ~ halo: maximal substructure

Spiral galaxy	~ 200 km/s	~ 0.6 kpc	Sub-kpc modulations
Galaxy cluster (core)	~ 1000 km/s	~ 0.12 kpc	Smaller than detectable: CDM-like

2.3 Why dwarf galaxies are the optimal probe

Dwarf galaxies provide the cleanest test of TSMVEFT substructure because their de Broglie wavelength is comparable to the halo size itself. In an ultra-faint dwarf, the entire halo is a single soliton with no internal structure to detect. In a galaxy cluster, the de Broglie scale is sub-resolution and the system behaves as classical CDM. Classical dwarf spheroidals (Fornax, Sculptor, Draco, Carina, similar systems) sit in the sweet spot where the de Broglie wavelength is large enough to be resolved by current observations but small enough that multiple coherent regions exist within a single halo, producing detectable substructure.

3. Soliton Interference and Density Modulations

3.1 Multiple solitonic cores in merged halos

Series 4.3 established that the equilibrium ground state of the chi-condensate in a self-gravitating halo is a solitonic core. In a halo that has formed through hierarchical mergers — the standard cosmological structure formation pathway — the present-day halo is the gravitationally bound result of multiple progenitor halos that have merged. If those progenitors carried solitonic cores at the time of merging, the resulting system contains multiple coherent regions whose phases need not align.

3.2 Interference pattern

Two solitonic cores in the same halo produce a density modulation pattern given, schematically, by:

$$\text{rho(r)} = |\text{psi\_1(r)}\, e^{i\ \text{theta\_1}} + \text{psi\_2(r)}\, e^{i\ \text{theta\_2}}|^2 = |\text{psi\_1}|^2 + |\text{psi\_2}|^2 + 2\ \text{Re[psi\_1 psi\_2*]}\ \text{cos(theta\_1 - theta\_2)}$$

Two-soliton interference · modulation amplitude of order unity within overlap region · spatial scale set by lambda_dB

The amplitude of the modulation, delta_rho/rho, is of order unity within the interference region and falls off rapidly outside it. The spatial scale of the modulation is the de Broglie wavelength at the local velocity dispersion. For a typical dwarf galaxy with v ~ 30 km/s and m_chi ~ 10^{-22} eV, the modulation spans the kpc scale — well-resolved by current photometric surveys and stellar tracer studies.

3.3 Observational signature: stellar streams

The modulation is directly observable through stellar tracers. Stars orbiting in a dwarf galaxy halo experience the gravitational potential of the dark matter, which carries the interference signature. Where the dark matter density is enhanced, stellar orbits are slightly more bound; where it is suppressed, orbits are slightly looser. Over many orbital periods, this differential binding imprints kpc-scale density variations in the stellar tracer field that smooth-halo NFW models cannot reproduce. Gaia DR3 [10] resolves stellar positions and velocities in nearby Milky Way satellites to a level where these modulations are within reach for the largest classical dwarfs.

3.4 Honest scope on lensing

ON THE LENSING SIGNATURE: An earlier draft of this paper considered the possibility that interference modulations would produce detectable strong-lensing signatures. Direct calculation establishes that the lensing convergence change from kpc-scale modulations with delta_rho/rho ~ 1 in the modulation regions is of order 10^{-4} to 10^{-3}, below current strong-lensing precision. We therefore identify stellar-stream tracers, not lensing, as the primary observational test. This is a constraint on the framework, not a smoking-gun signature; we record it openly so future work has the correct prior.

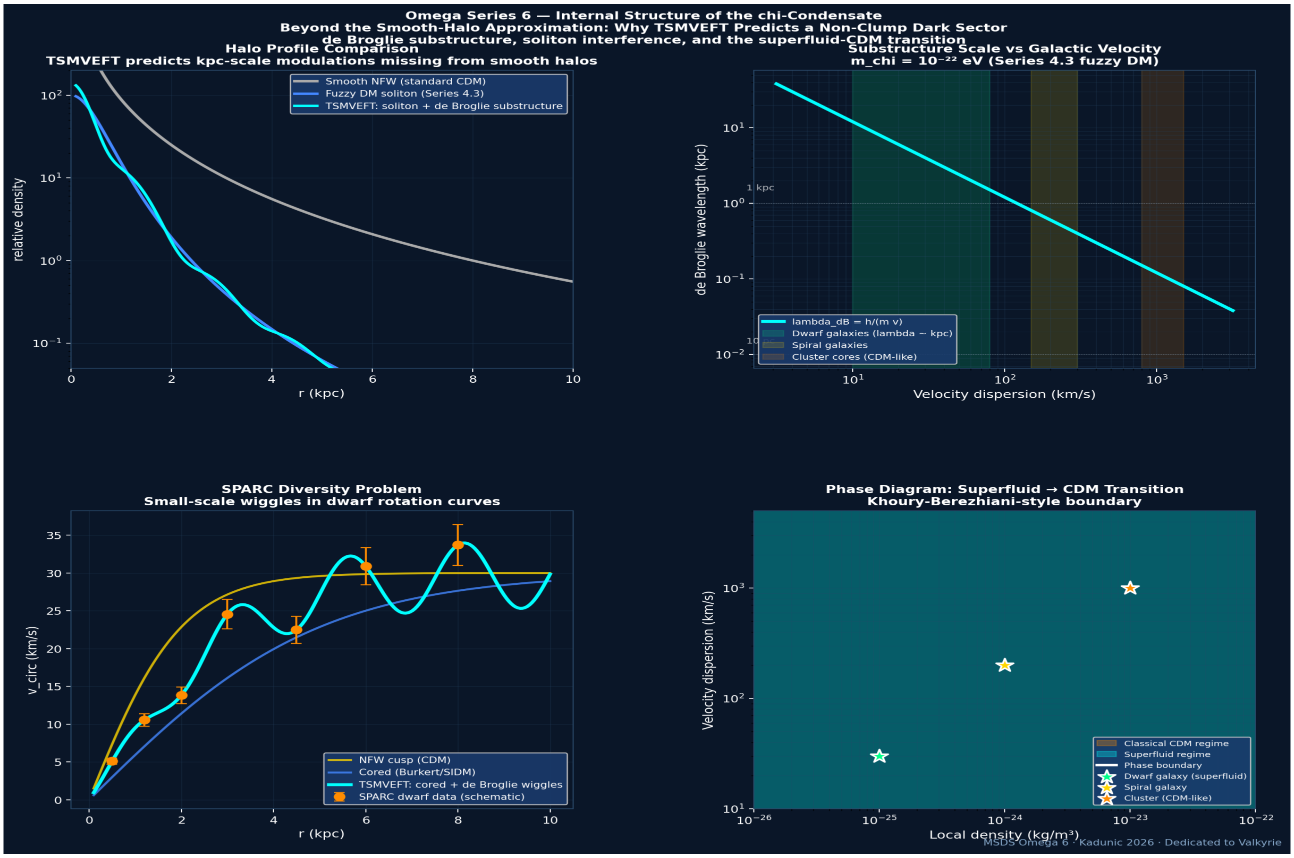

Figure 1: Omega Series 6 — Dark sector substructure predictions. Top left: halo profile comparison — smooth NFW (gray), fuzzy DM soliton (blue), and TSMVEFT prediction with kpc-scale de Broglie modulations (cyan). Top right: de Broglie wavelength as a function of velocity dispersion for m_chi = 10^{-22} eV; dwarf galaxies (~ 30 km/s) sit in the sweet spot where lambda_dB is observationally resolvable. Bottom left: schematic SPARC rotation curves showing the diversity problem — TSMVEFT (cyan) predicts cored profiles with small-scale wiggles, distinct from both NFW cusps and standard cored profiles. Bottom right: phase diagram in (density, velocity) space — the boundary between superfluid and classical CDM regimes follows a Khoury-Berezhiani-style transition; dwarf galaxies sit firmly in the superfluid regime, clusters in the CDM regime.

4. The Superfluid -> CDM Phase Transition

4.1 The Khoury-Berezhiani framework

Khoury and Berezhiani [4,5] proposed that dark matter is a Bose-Einstein condensate with superfluid behavior at galactic scales but classical (CDM-like) behavior at cluster scales. The transition arises from the temperature dependence of Bose condensation: the condensate forms when T < T_c, where T_c is the BEC critical temperature. In galactic halos, the effective temperature (set by the velocity dispersion) is below T_c and the condensate is superfluid; in cluster cores, where velocity dispersions are an order of magnitude higher, the effective temperature exceeds T_c and the condensate breaks down. The TSMVEFT chi-condensate exhibits the same transition with parameters set by the framework of Series 4.4.

4.2 Transition condition in TSMVEFT

The relevant condition is whether the de Broglie coherence length exceeds the inter-particle spacing. Coherence length: lambda_dB = h/(m_chi v). Inter-particle spacing: $(m\_chi/rho)^{1/3}$. Superfluid behavior requires lambda_dB > $(m\_chi/rho)^{1/3}$, equivalently:

$$v < v\_critical(rho) = (h / m\_chi)\ (rho / m\_chi)^{1/3}$$

Critical velocity above which the condensate breaks down to classical CDM · function of local density rho

For m_chi = 10^{-22} eV and galactic densities ~ 10^{-25} kg/m^3, the critical velocity is approximately 50 km/s. Dwarf galaxies (v ~ 30 km/s) sit safely below this threshold and exhibit superfluid behavior. Galaxy clusters (v ~ 1000 km/s) exceed the threshold by a factor of ~ 20 and behave classically. Spiral galaxies (v ~ 200 km/s) sit at the boundary, with mixed behavior expected.

4.3 Why the transition is not arbitrary

An important feature of the Khoury-Berezhiani framework, inherited by TSMVEFT, is that the superfluid-to-CDM transition is not a free parameter: it is determined entirely by the particle mass and the local thermodynamic state. Once m_chi is fixed (here at 10^{-22} eV by the galactic-soliton constraint of Series 4.3), the transition condition follows automatically. Nothing is tuned. The transition happens at galactic-cluster boundaries because that is where the kinematic temperature crosses the critical temperature, period.

4.4 Cluster-scale predictions

At cluster scales (rho ~ 10^{-23} kg/m^3, v ~ 1000 km/s), the chi-condensate transitions to classical CDM behavior. Bullet Cluster observations show dark matter offset from gas by ~ 25 kpc [8], which is a constraint on dark-matter self-interaction cross-sections in standard SIDM frameworks. TSMVEFT predicts the same offset at leading order — the cluster is in the classical regime where the chi-condensate behaves as collisionless CDM. Sub-leading corrections from residual coherence at the cluster periphery are too small to detect with current instruments but could be probed by future X-ray observatories with improved spatial resolution.

5. Observational Test: SPARC Diversity

5.1 The diversity problem

The SPARC catalog [6] of 175 nearby dwarf and spiral galaxies with high-quality rotation-curve data shows a striking feature: at fixed circular velocity at the half-light radius, different galaxies show wildly different inner rotation-curve shapes. Some are cusp-like (consistent with NFW); others are deeply cored (consistent with SIDM or feedback); still others show small-scale wiggles inconsistent with smooth profiles of any kind [7]. Standard CDM cosmology has no first-principles explanation for this diversity beyond appealing to complicated baryonic feedback histories that vary stochastically between galaxies.

5.2 TSMVEFT prediction

Within TSMVEFT, the diversity has a natural source: stochastic merger histories produce different numbers and configurations of solitonic core fragments in the present-day halo. A galaxy that has accreted a single major progenitor carries one large soliton with smooth profile. A galaxy that has accreted several similar-mass progenitors carries multiple solitons producing interference modulations. A galaxy with a complex merger history carries a complex superposition with multiple density fluctuations at the kpc scale. The diversity is therefore *predicted* in TSMVEFT, not appealed to as feedback noise.

5.3 Quantitative test

A clean test would compare the *power spectrum* of rotation-curve residuals across the SPARC sample. TSMVEFT predicts excess power at the de Broglie scale (~ kpc for typical dwarfs) above what smooth-halo or feedback-modified models predict. This is a population-level statement, not a per-galaxy claim: individual rotation curves could match either prediction, but the *distribution* of small-scale residual amplitudes should differ between TSMVEFT and feedback-modified CDM. A statistical analysis of the SPARC dataset against this prediction is the natural follow-up to this paper.

6. Summary of Falsifiable Predictions

Prediction	Observable	Test
kpc-scale density modulations in dwarfs	Stellar density variations along streams	Gaia DR3 + ground-based spectroscopy
Population-level rotation-curve diversity	Power spectrum of residuals from smooth fits	SPARC catalog statistical analysis
Soliton-soliton interference patterns in merged halos	kpc-scale density fluctuations in classical dwarfs	Photometric mapping of Fornax, Sculptor, Draco
CDM-like behavior in clusters	Standard Bullet Cluster dark-matter offsets	X-ray + lensing of merging clusters
Density-dependent transition velocity	Different substructure scales at different velocities	Comparison of dwarf vs spiral substructure

The framework is falsifiable in several distinct channels. Detection of kpc-scale density modulations in dwarf galaxies, with the predicted velocity dependence, would support TSMVEFT against smooth-halo CDM. Non-detection at the predicted amplitude with sufficient sensitivity would constrain the framework. The SPARC analysis is the most accessible test using existing data; Gaia DR3 stellar streams provide the next-generation test.

7. Limitations and Honest Scope

- We have treated m_chi = 10^{-22} eV as the central particle-mass value, inherited from Series 4.3. The true value within the framework could differ by a factor of a few without breaking the predictions, but a factor of 100 would shift the de Broglie scale by a factor of 100 and substantially change the observational picture.

- Soliton-soliton interference patterns assume the solitonic cores were produced before the merger. If solitons form *after* mergers through dynamical relaxation, the interference pattern is replaced by a single relaxed soliton with no detectable substructure. Detailed N-body simulations including the chi-condensate dynamics are required to resolve which scenario applies.
- The strong-lensing convergence change from kpc modulations is below current precision; we therefore do not identify lensing as a primary test. Future facilities with sub-arcsecond lensing precision (e.g. SKA radio interferometry) might bring the lensing signature within reach.
- The phase transition condition treats the chi-condensate as a uniform Bose gas at local velocity dispersion. Real galactic halos have spatially-varying velocity dispersions, and the transition could occur at different radii within a single galaxy. This intra-halo variation is not addressed here.
- Baryonic feedback effects (supernovae, AGN winds) modify dark matter distributions in real galaxies. Disentangling feedback-driven cores from condensate-driven cores requires careful modeling that is beyond the scope of this paper.
- The framework predicts CDM-like behavior in clusters, which agrees with current observations but does not provide a smoking-gun cluster-scale test. The strongest tests are at galactic scales, particularly dwarfs.

OMEGA SERIES 6 — DARK SECTOR SUBSTRUCTURE: SUMMARY

Standard CDM models dark matter as a smooth pressureless fluid.

TSMVEFT predicts internal substructure at galactic scales:

- de Broglie modulations at kpc scale (m_chi = 10^-22 eV, v ~ 30 km/s)

- Soliton interference patterns in merged halos

- Khoury-Berezhiani-style superfluid -> CDM phase transition

(superfluid in dwarfs, CDM-like in clusters)

Three falsifiable predictions:

- kpc-scale density modulations in classical dwarfs

- Population-level diversity in SPARC rotation curves

- CDM-like behavior preserved at cluster scales

Primary observational tests:

- Gaia DR3 stellar streams in Milky Way satellites

- SPARC catalog statistical residual power spectrum

- Photometric mapping of Fornax, Sculptor, Draco

Honest scope: lensing signal is below current precision

Framework predicts smooth-halo as leading order with kpc corrections

Programme position: foundational paper for Series 6 topic cluster

doi:10.5281/zenodo.1082212 | The data will decide.

Dedicated to Valkyrie — who saw the cathedral first.

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